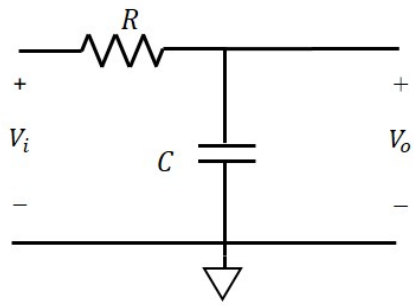
Lab 7 - Transient Response

Objective

The goal of this lab is to explore the transient behavior of a 1st order circuit. To do so, you will build a circuit that will cause an LED to flash at a frequency of your choosing.

Introduction



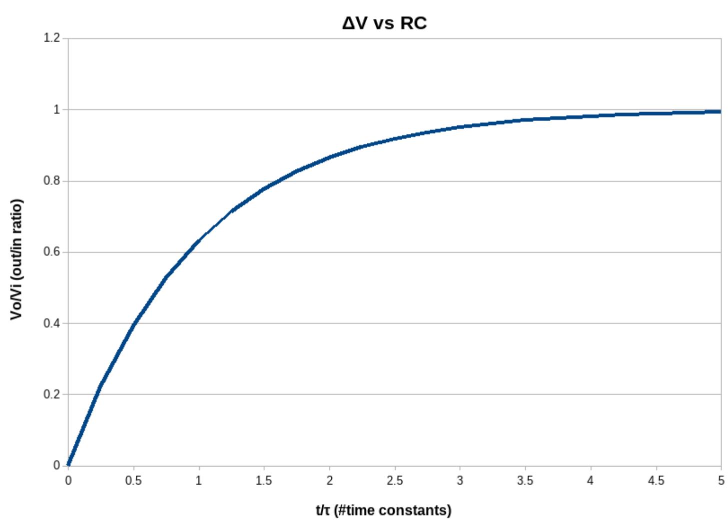
In the RC circuit shown above, a resistor and a capacitor form a first order circuit. Based upon the previous lecture, we can describe this circuit using the following differential equation:

$$C\frac{dV_o}{dt}+\frac{V_o-V_i}{R}=0 \Rightarrow \frac{dV_o}{dt}+\frac{V_o}{RC}=\frac{V_i}{RC}$$

if V_i is a constant and $V_o(0)=V_o$, then

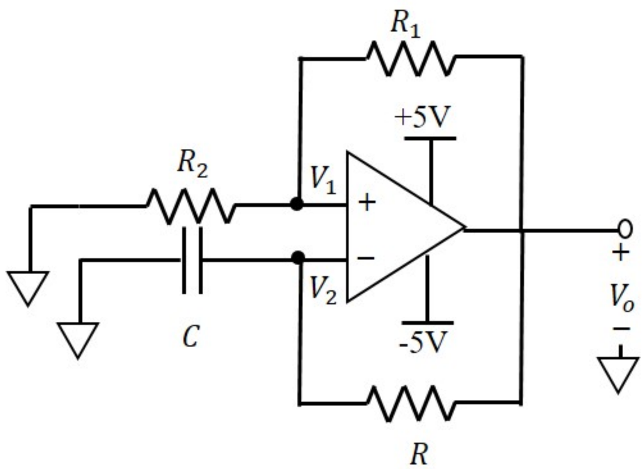
$$V_o(t)=V_i+(V_o-V_i)e^{-\frac{t}{RC}}$$

Thus, the voltage transitions from a starting value of to a final value of in an exponential fashion. The RC time constant (τ) determines the rate at which the output voltage transitions from its starting to ending values. Note, by the time t=5τ, the output voltage has very nearly reached its final value (see graph below, with $V_o(0)=0$).



Prelab

In this lab, we will use an an op-amp to to blink LEDs based at a certain frequency that is based upon the RC time constant.



Using LTSpice, simulate the circuit above. Note, you will be using a different op-amp than in previous labs. Specifically, you will need to use the “UniversalOpAmp2” instead of the generic opamp we have been using. In addition, you will need to add +Vcc and -Vcc to power this op-amp as well (add 2 voltage sources attached to ground, but with opposite ends, and make the other end +Vcc/-Vcc - you can then use these labels directly on to the voltage inputs on the op-amp).

As the “feedback” is connected via an RC circuit, we must analyze this a little differently than our previous, resistor-only, op-amp circuits. Looking at the non-inverting input, we know that the current flowing into the op-amp must be zero (or nearly zero). As such, we can determine the V_o using a voltage divider, specifically:

$$\frac{V_1-0}{R_2}+\frac{V_1-V_0}{R_1}=0\Rightarrow V_1\left(\frac{1}{R_1}+\frac{1}{R_2}\right)=\frac{V_0}{R_1}\Rightarrow V_1=V_o\frac{R_2}{R_1+R_2}=\gamma V_o$$

where $\gamma=\frac{R_2}{R_1+R_2}$ Similarly, looking at the inverting input, and following a similar logic (current into op-amp must be zero), the equation is:

$$C\frac{dV_2}{dt}+\frac{V_2-V_o}{R}=0\Rightarrow\frac{dV_2}{dt}+\frac{V_2}{RC}=\frac{V_o}{RC}$$

We now see that V_1 and V_2 are related to V_o in very different ways. As such, we expect that V1 ≠ V2. Since these are the inputs to the op-amp and whenever these inputs are NOT equal, the op-amp will saturate. Thus, we can conclude that, in this circuit, the op-amp will be running in saturation mode. As such, we need to consider two cases: $V_2 < V_1$, and vice-versa.

Case 1: $V_2 < V_1$

In this case, V_1 will be constant and equal to γV_{sat}, thus:

$$\frac{dV_2}{dt}+\frac{V_2}{RC}=\frac{V_{sat}}{RC}\Rightarrow V_2(t)=V_{sat}+(V_2(0)-V_{sat})e^{-\frac{t}{RC}}$$

which is similar to the behavior discussed in the introduction. Now, as V_2 continues to grow toward its max (V_{sat}), it will become larger than V_1 (which is less that V_{sat}). This leads us to case 2.

Case 2: $V_1 < V_2$

In this case, V_1 will be constant and equal to -γV_{sat}, and V_2 will follow:

$$V_2(t)=-V_{sat}+(V_2(0)+V_{sat})e^{-\frac{t}{RC}}$$

Now, as V_2 continues to shrink (grow "negatively") toward its min (-V_{sat}), it will eventually become lower than V_1 (which is greater that -V_{sat}), eventually taking us back to case 1.

Thus, this circuit will oscillate between these two cases. As the voltages are all periodic, they will behave similar to the behavior outlined in the introduction. We can determine the periodicity by solving for the time by letting t=0 and $V_2(0)$ = -γV_{sat}. The first "switch" will occur when V(t) = γV_{sat}, namely solving for t in the following:

$$V_2(t)=V_{sat}+(-\gamma V_{sat}-V_{sat})e^{-\frac{t}{RC}}=\gamma V_{sat}$$

After algebraic manipulation, we find that:

$$t=-\tau\ln\left(\frac{1-\gamma}{1+\gamma}\right)$$

But this only represents half of the periodicity ($V_2<V_1$); the total period is 2X this value. As the frequency is 1/t, and this is only dependent upon γ, which is only dependent upon R_1 and R_2, we can pick these resistor values to set the frequency of the LED on/off cycle.

Simulations

For your LTSpice circuit, select values of R_1, R_2, R, and C such that:

- frequency is ~1Hz (RC=1)
- R1 + R2 ≈ 10kΩ (=R)

When you're ready to run your simulation, you will need to add the following initial condition as a SPICE directive: `.ic V(Vo) = 1`. Then, when you hit RUN, you will need to switch to the "Transient" tab in the popup and enter a stop time, typically 3-5X of your period which, in this case, should be 1 second. After you hit OK, a second window will appear showing the waveforms, but they will be blank. You need to tell LTSpice what to measure. If you "hover" over your circuit, a probe icon will appear. You can click it on various points and those will be plotted on your waveform window. Please submit your LTSpice circuit and waveform image with your lab report.

Procedure

Parts needed:

- A selection of 1/4W resistors
- A selection of capacitors
- One 741 (or equivalent) op-amp
- One red LED
- One green LED
- One 10kΩ potentiometer w/turner (slotted screwdriver)
- A selection of colored 22 gauge connection wires

It is a good idea to always verify your component values before you begin doing an experiment. This will allow you to identify one possible source of error easily.

Build and Measurements (Task 1)

Build the circuit shown above in the prelab section. If you have selected your correctly, the two LEDs should flash on/off, alternatively, approximately once per second. Use an oscilloscope and measure the following for both $V_1(t)$ and $V_2(t)$:

- Actual frequency of oscillation
- Peak-to-Peak Voltage
- Root-Mean-Square (RMS) Voltage

Note any differences between your measurements and your simulation and explain these in your lab report.

Build and Measurements (Task 2)

For this circuit, we will replace R_1 and R_2 with a potentiometer. A potentiometer is a 3-pin device that "splits" the total resistance into two parts, the sum of which is the total. As such, it functions as a variable voltage divider. Note, if you leave one of the end terminals disconnected, the pot acts like a variable resistor (varistor).

- Step 1
  - Replace R_1 and R_2 with the 10kΩ potentiometer. Adjust the pot until your circuit oscillates at a frequency of 2Hz. Take a picture or drawing of your waveform for your lab report. Remove the pot from the circuit and measure/record its values for R_1 and R_2. What is the corresponding gamma?
- Step 2
  - Replace R (from the RC side) with half its value. Re-insert the pot back into the circuit and adjust its value until the circuit oscillates at 2Hz again. Take a picture or drawing of your waveform for your lab report. Remove/measure/record its values for R_1 and R_2. What is the corresponding gamma?

Lab Report

- Be sure to have a title page with the usual items (e.g. lab name, your name, date).
- Prelab design and results
- Procedure – Write in your own words what you did in lab
- Measured Data – Include tables of values you measured throughout the lab.
- Measured Waveforms – Be sure that you label your images and intersperse them (with comments) through your document
- Discussion –Be sure to answer any questions posted in the procedure section. Also, discuss differences between theoretical values calculated in the prelab and actual values measured in the lab. Definitely comment on what you think are the major sources of any significant differences.